1. Student Grade Calculator

A college wants to automatically assign grades based on a student's marks:

- 90–100 → A+
- 80–89 → A
- 70–79 → B
- 60–69 → C
- 50–59 → D
- Below 50 → Fail

Accept the student's mark and display the appropriate grade.

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2. ATM Withdrawal System

An ATM should process a withdrawal request.

- If the requested amount is greater than the account balance → display **Insufficient Balance**
- If the amount is not a multiple of ₹100 → display **Enter an amount in multiples of ₹100**
- Otherwise → display **Withdrawal Successful**

Accept the account balance and withdrawal amount.

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3. Online Shopping Delivery Charge

An e-commerce website calculates delivery charges based on the order value:

- ₹5,000 or more → Free delivery
- ₹2,000–₹4,999 → ₹50 delivery charge
- Below ₹2,000 → ₹100 delivery charge

Calculate and display the delivery charge and final payable amount.

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4. Employee Performance Rating

A company rates employees based on their performance score:

- 90 or above → Excellent
- 75–89 → Very Good
- 60–74 → Good
- 40–59 → Needs Improvement
- Below 40 → Poor

Accept the performance score and display the rating.

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5. Electricity Bill Calculator

An electricity provider charges according to usage:

- First 100 units → ₹2/unit
- Next 100 units → ₹3/unit
- Next 200 units → ₹5/unit
- Above 400 units → ₹7/unit

Accept the number of units consumed and calculate the total electricity bill.

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6. Movie Ticket Pricing

A cinema determines ticket prices based on age:

- Below 5 → Free
- 5–12 → ₹100
- 13–59 → ₹200
- 60 or above → ₹120

Accept the customer's age and display the ticket price.

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7. Traffic Fine System

A traffic department calculates fines based on vehicle speed.

- Up to 60 km/h → No fine
- 61–80 km/h → ₹500

- 81–100 km/h → ₹1,000
- Above 100 km/h → ₹2,000 and License Review

Accept the vehicle speed and display the appropriate result.

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8. Restaurant Billing System

A restaurant provides discounts based on the total bill:

- ₹5,000 or above → 20% discount
- ₹3,000–₹4,999 → 15% discount
- ₹1,000–₹2,999 → 10% discount
- Below ₹1,000 → No discount

Calculate the discount and final bill amount.

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9. College Admission Decision

A college evaluates applicants based on their entrance examination score.

- 90+ → Direct Admission
- 75–89 → Admission + Interview
- 60–74 → Waitlisted
- Below 60 → Not Eligible

Accept the entrance score and display the admission status.

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10. Bank Loan Risk Category

A bank categorizes applicants based on their credit score:

- 750 or above → Low Risk
- 650–749 → Medium Risk
- 550–649 → High Risk
- Below 550 → Very High Risk

Additionally, if the applicant has an existing loan default, display **Loan Approval Requires Manual Review**, regardless of the score.

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11. Hotel Room Pricing

A hotel charges different prices depending on the room type:

- Standard → ₹2,000/night
- Deluxe → ₹3,500/night
- Suite → ₹5,000/night

If the customer stays for more than 5 nights, provide a 10% discount on the total room cost.

Accept room type and number of nights and calculate the final bill.

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12. Employee Bonus Calculator

A company calculates annual bonuses based on performance:

- Performance ≥ 90 → 20% of annual salary
- Performance 75–89 → 15%
- Performance 60–74 → 10%
- Below 60 → No bonus

Accept annual salary and performance score and calculate the bonus.

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13. Mobile Data Plan

A telecom provider recommends a plan based on monthly data usage:

- Below 5 GB → Basic Plan – ₹199
- 5–20 GB → Standard Plan – ₹399
- 21–50 GB → Premium Plan – ₹699
- Above 50 GB → Unlimited Plan – ₹999

Accept the customer's monthly data requirement and recommend a plan.

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14. Parking Fee Calculator

A parking facility charges according to parking duration:

- Up to 2 hours → ₹30
- 3–5 hours → ₹50
- 6–10 hours → ₹100
- More than 10 hours → ₹150

If the vehicle is an electric vehicle, apply a 20% discount.

Calculate the final parking fee.

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15. Online Exam Result

A student must satisfy **both** conditions to pass:

- Theory mark ≥ 40
- Practical mark ≥ 40

If both are passed, calculate the average:

- Average ≥ 75 → Distinction
- Average ≥ 60 → First Class
- Average ≥ 50 → Second Class
- Otherwise → Pass

If either subject is below 40, display **Fail**.

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16. Flight Baggage Fee

An airline allows 15 kg of free baggage.

- Up to 15 kg → No extra charge
- 16–20 kg → ₹500
- 21–30 kg → ₹1,000
- Above 30 kg → ₹2,000

If the passenger is travelling in Business Class, the free baggage limit is 25 kg.

Calculate the applicable baggage charge.

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17. E-Commerce Return Eligibility

An online store evaluates return requests.

A product can be returned if:

- It was purchased within 7 days,
- The product is unused,
- The product is returnable.

If all conditions are satisfied → **Return Approved**.

Otherwise, display the appropriate reason for rejection.

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18. Salary Tax Calculator

A simplified tax system uses the following annual income slabs:

- Up to ₹3 lakh → No tax
- ₹3–6 lakh → 5%
- ₹6–10 lakh → 10%
- Above ₹10 lakh → 20%

Accept the annual salary and calculate the tax amount and final salary after tax.

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19. Ride Fare Calculator

A ride-sharing application calculates fares based on distance:

- First 5 km → ₹100
- 6–15 km → ₹15/km
- 16–30 km → ₹12/km
- Above 30 km → ₹10/km

If the ride is booked during peak hours, an additional 20% surcharge is applied.

Calculate the final fare.

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20. Employee Leave Approval System

An employee requests leave from a company.

The request should be evaluated as follows:

- If requested days are greater than available leave → **Leave Rejected**
- If available leave is sufficient and requested days ≤ 3 → **Leave Automatically Approved**
- If requested days are 4–7 → **Manager Approval Required**
- More than 7 days → **HR Approval Required**

Accept available leave and requested leave and display the appropriate decision.